Math 340: Elementary Matrix and Linear Algebra

MIDTERM TWO

Thursday, November 14th, 2024, 7:30pm-9:00pm

Circle your Instructor and your TA:

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Name: ____________________________________ Wisc email: ______________________________

I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.

Student Signature: ____

READ THE FOLLOWING INFORMATION.

- This is a 90-minute exam. It consists of eight problems for a total of 50 points; the exam is six sheets of paper, including this cover sheet. It is your responsibility to make sure that you have a complete exam.
- Books, notes, calculators, and other aids are not allowed.
- Problems are spaced out to allow ample room for work. You may use the last page for scratch work, but it will not be graded. Do not unstaple or remove pages as they can be lost in the grading process. **An incomplete exam packet will result in an automatic zero.**
- Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.
- If making multiple attempts at a solution, indicate clearly which attempt you'd like to be graded by crossing out the other answers. Multiple attempts will not be graded.
- Notation reminders: M_{mn} is the vector space of $m \times n$ real matrices with standard matrix addition and scalar multiplication, P_d is the vector space of polynomials (with single variable x) of degree at most d with standard polynomial addition and scalar multiplication, $\mathbf{0}$ denotes the zero vector in a vector space V. $\mathbb{R}^n$ uses the standard vector addition/scalar multiplication unless otherwise noted.
- Additional notation reminders: O denotes the zero matrix, and I_n denotes the $n \times n$ identity matrix. $\operatorname{tr}(A)$ denotes the trace of a matrix A.

DO NOT BEGIN THIS EXAM UNTIL SIGNALED TO DO SO.

1. (8 points, 1 point each) Clearly mark the correct answer(s) for each of the following by completely filling in the appropriate bubble. **No justification is needed.**

   a.) Let A be an $n \times m$ matrix. If $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a set of linearly independent vectors in $\mathbb{R}^m$, then $\{A\mathbf{v}_1, \dots, A\mathbf{v}_n\}$ is a set of linearly independent vectors in $\mathbb{R}^n$.

   ◯ True
   ◯ False

   b.) For the matrix $A = \begin{bmatrix} 1 & -1 \\ 3 & -3 \end{bmatrix}$, the image of A, $\text{im}A$, is a subspace of $\mathbb{R}^2$ with $\dim(\text{im}A) = 2$.

   ◯ True
   ◯ False

   c.) If A is 5×5 diagonalizable matrix, then A must have 5 distinct eigenvalues.

   ◯ True
   ◯ False

   d.) Let V be a vector space, and $W = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ be a subspace of V. If $\mathbf{v}$ is a vector in V which is not in W, then $\mathbf{v}$ is not a linear combination of $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$.

   ◯ True
   ◯ False

   e.) In P_2, the vectors $4x^2 - 2x + 1$ and $2x^2 - x + 3$ are linearly dependent.

   ◯ True
   ◯ False

f.) **(Multiple Choice-choose one)** Suppose A is a 2×2 matrix where $AP = PD$ for $P = \begin{bmatrix} 1 & 3 \\ 1 & 2 \end{bmatrix}$ and $D = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$. Note that $P^{-1} = \begin{bmatrix} -2 & 3 \\ 1 & -1 \end{bmatrix}$. What is A^{10}?

- ◯ $\begin{bmatrix} -2a^{10}+3b^{10} & 3a^{10}-3b^{10} \\ -2a^{10}+2b^{10} & 3a^{10}-2b^{10} \end{bmatrix}$
- ◯ $\begin{bmatrix} -2a^{10}+3b^{10} & -6a^{10}+6b^{10} \\ a^{10}-b^{10} & 3a^{10}-2b^{10} \end{bmatrix}$
- ◯ $\begin{bmatrix} a^{10} & 0 \\ 0 & b^{10} \end{bmatrix}$
- ◯ None of the above.

g.) **(Multiple Choice-choose one)** Consider a set $S = \{p_1, p_2, p_3, p_4, p_5\}$ of 5 polynomials in P_3. Which of the following is true?

- ◯ S spans P_3.
- ◯ S is a linearly independent set.
- ◯ S is a basis for P_3.
- ◯ At least one of the polynomials in S is a linear combination of the others.
- ◯ None of the above.

h.) **(Multiple Choice-choose one)** For what value(s) of c do the vectors $\begin{bmatrix} 3 \\ 0 \\ c \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$ form a basis for $\mathbb{R}^3$?

- ◯ $c = 9$
- ◯ All values of c except $c = 9$
- ◯ No values of c make this a basis.
- ◯ All values of c make this a basis.
- ◯ None of the above.

2. (4 points total, 1 point each) Suppose A is a 6×4 matrix of rank 4, and consider the linear transformation $T : \mathbb{R}^4 \to \mathbb{R}^6$ by $T(\mathbf{x}) = A\mathbf{x}$. Answer each of the following questions. Justification is not required.

   a. What is the dimension of the kernel of T? Fill in the blank below.

   $\dim \ker(T) = \boxed{}$

   b. **(Multiple Choice-choose one)** Is T one-to-one?

   - ◯ Yes
   - ◯ No
   - ◯ Not enough information to answer.

   c. What is the dimension of the image of T? Fill in the blank below.

   $\dim \operatorname{im}(T) = \boxed{}$

   d. **(Multiple Choice-choose one)** Is T onto?

   - ◯ Yes
   - ◯ No
   - ◯ Not enough information to answer.

3. (5 points)

   Suppose $T : P_1 \to M_{22}$ is a linear transformation with $T(4x+3) = \begin{bmatrix} 1 & 3 \\ -2 & 2 \end{bmatrix}, T(x+1) = \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix}$.
   Find $T(1)$. Show all work.

4. (6 points total) Consider $A = \begin{bmatrix} 2 & -2 \\ 0 & 1 \end{bmatrix}$, and let $W = \{X \text{ in } M_{22} \mid AX = XA\}$.

   a. (1 point) Show that W is nonempty.

   b. (3 points) Is W closed under addition? Fully justify your answer using complete sentences.

   c. (2 points) Is W closed under scalar multiplication? Fully justify your answer using complete sentences.

5. (11 points total) Let V be $\mathbb{R}^2$ with the following vector addition $\oplus$ and scalar multiplication $\odot$:

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \oplus \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 + 4 \\ y_1 + y_2 - 5 \end{bmatrix}, \quad c \odot \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} cx_1 + 4c - 4 \\ cy_1 - 5c + 5 \end{bmatrix}$$

a.) (2 point) Compute $\begin{bmatrix} 2 \\ 4 \end{bmatrix} \oplus \begin{bmatrix} 0 \\ -1 \end{bmatrix}$.

b.) (3 points) Find the zero vector $\mathbf{0}$ for this vector addition $\oplus$. You must justify that $\mathbf{v} \oplus \mathbf{0} = \mathbf{v} = \mathbf{0} \oplus \mathbf{v}$ for all $\mathbf{v}$ in $\mathbb{R}^2$.

c.) (3 points) For $\mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix}$, find the additive inverse of $\mathbf{v}$ (i.e. find $\mathbf{u}$ such that $\mathbf{u} \oplus \mathbf{v} = \mathbf{0} = \mathbf{v} \oplus \mathbf{u}$). Fully justify your answer.

d.) (3 points) Show the following vector space property holds: $1 \odot \mathbf{u} = \mathbf{u}$ for all $\mathbf{u}$ in $\mathbb{R}^2$.

6. (6 points total) For $A = \begin{bmatrix} 10 & 6 & 12 \\ -3 & 1 & -6 \\ -3 & -3 & -2 \end{bmatrix}$:

   a.) (4 points) Find a basis for the eigenspace associated to the eigenvalue $\lambda = 4$ (in other words, find a set of basic eigenvectors associated to $\lambda = 4$). Show all work.

   b.) (2 points) Given that $\lambda = 1$ is the only other eigenvalue of A, and a basis for the eigenspace of $\lambda = 1$ is $\left\{ \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \right\}$, answer the following: Is A diagonalizable? Fully justify your answer using complete sentences.

7. (4 points total, 1 point each) For this problem, let A be a 3×5 matrix whose reduced row echelon form $RREF(A)$ is given as $RREF(A) = \begin{bmatrix} 1 & -2 & 0 & 3 & 0 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$. Note: This is the **reduced row echelon form of** A, not A. Answer each of the following questions about the matrix A, if possible. Justification is not required.

   a. **(Multiple Choice-choose one)** What is the rank of A?

      ◯ 1
      ◯ 2
      ◯ 3
      ◯ 4
      ◯ 5
      ◯ Not enough information to answer.

   b. **(Multiple Choice-choose one)** For which value of c is $\begin{bmatrix} c \\ 0 \\ -1 \\ 1 \\ 0 \end{bmatrix}$ in the nullspace of A?

      ◯ $c = -3$
      ◯ $c = 0$
      ◯ $c = 1$
      ◯ $c = 3$
      ◯ Not enough information to answer.

   c. **(Multiple Choice-choose one)** Is $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ in the column space of A?

      ◯ Yes
      ◯ No
      ◯ Not enough information to answer.

   d. **(Multiple Choice-choose one)** Are the (five) column vectors making up A a linearly independent set in $\mathbb{R}^3$?

      ◯ Yes
      ◯ No
      ◯ Not enough information to answer.

8. (6 points total) Each of the following statements is **false**. Show each statement is false by providing explicit counterexamples. You must fully justify your answers.

   a. (2 points) The function $F : M_{22} \to \mathbb{R}$ by $F(A) = (\mathrm{tr}(A))^2$ is a linear transformation.

   b. (2 points) The set of vectors $W = \left\{ \begin{bmatrix} a \\ b \end{bmatrix} \middle| \, a+b \geq -1 \right\}$ is a subspace of $\mathbb{R}^2$.

   c. (2 points) Let V be $\mathbb{R}^2$ with the operation $\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \oplus \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ y_1 \end{bmatrix}$. $\oplus$ satisfies the vector space property $\mathbf{v} \oplus \mathbf{u} = \mathbf{u} \oplus \mathbf{v}$.

Scratch paper page!!

Back of scratch paper page!